SUPPLY RISK ASSESSMENT REPORT — STRATEGIC RAW MATERIALS

Reference: SDG-CRM-2025-009
Classification: INTERNAL USE ONLY — RESTRICTED DISTRIBUTION
Prepared by: Strategic Supply Directorate — Risk Analysis Unit
Reviewed by: Chief Operating Officer; Legal Affairs Office — Trade & Procurement Division
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Next scheduled review: August 2025

EXECUTIVE SUMMARY

This report presents the findings of the Strategic Supply Directorate's biannual supply risk assessment for the organization's portfolio of strategic raw materials, conducted in accordance with the internal Risk Assessment Protocol (RAP v4.1) and in compliance with the monitoring and reporting obligations established under Regulation (EU) 2024/1252 on critical raw materials (the Critical Raw Materials Act, hereinafter CRMA). The assessment covers four materials classified as strategic under Annex I of the CRMA: battery-grade lithium carbonate, neodymium, cobalt sulphate, and spherical anode-grade natural graphite. Together, these materials represent approximately 34% of the organization's total direct material input by value and are incorporated in finished products supplied to customers operating in the clean energy, advanced manufacturing, and defense electronics sectors.

The assessment concludes that the organization's overall supply risk exposure has increased moderately compared to the previous assessment period (Q2–Q3 2024). The principal drivers of this increase are: continued high supply concentration for neodymium and natural graphite in a single third-country jurisdiction; quality consistency issues with the incumbent cobalt sulphate supplier; and the imminent expiry of the primary lithium carbonate framework agreement. Two materials — neodymium and natural graphite — are assessed as HIGH risk for the current period. Lithium carbonate is assessed as MEDIUM risk, primarily due to the contract expiry timeline. Cobalt sulphate is assessed as MEDIUM-LOW risk, with the quality issue noted as a watch item.

The report concludes with a set of twelve recommendations addressed to the Critical Resources Management Committee, covering procurement strategy adjustments, supplier qualification activities, and regulatory compliance actions required before the Q2 2025 reporting deadline under Article 20 of the CRMA.

1. Scope, Methodology, and Regulatory Framework

1.1 Scope of the Assessment

This assessment covers the four strategic raw materials identified in the organization's 2025–2027 Strategic Raw Materials Plan as Tier 1 procurement priorities. Tier 1 designation reflects a combination of the following criteria: annual procurement value above EUR 2 million; incorporation in products subject to supply continuity obligations with key customers; classification as strategic under Annex I of the CRMA; and absence of qualified technical substitutes within a 12-month substitution horizon. Materials classified as critical but not strategic under the CRMA, and materials for which qualified substitutes are available, are assessed separately under the Tier 2 assessment process and are not addressed in this report.

The assessment does not cover indirect supply chain exposure beyond Tier 1 suppliers. The Directorate acknowledges that full supply chain transparency to extraction level is not currently achievable for all materials in scope, and that the supply risk ratings assigned in this report therefore reflect primarily the organization's direct procurement exposure. Efforts to extend supply chain traceability to upstream processing and extraction stages are ongoing and will be reported separately under the CRMA Article 20 compliance submission due in April 2025.

1.2 Methodology

The supply risk assessment applies the organization's standard Risk Assessment Protocol (RAP v4.1), which evaluates each material across five risk dimensions: supply concentration (SC), price volatility (PV), regulatory and trade policy risk (RT), supplier financial and operational stability (SF), and substitutability (SU). Each dimension is scored on a scale of 1 to 5, where 1 represents minimal risk and 5 represents critical risk. The composite risk score is calculated as a weighted average of the five dimension scores, with weights assigned based on the material's supply chain characteristics and the organization's strategic exposure. The resulting composite score maps to a four-level risk classification: LOW (composite ≤ 2.0), MEDIUM-LOW (2.1–3.0), MEDIUM (3.1–3.9), HIGH (≥ 4.0).

The assessment draws on the following data sources: the European Commission's Q4 2024 supply chain stress test conducted under Article 19 of the CRMA; pricing data from Fastmarkets and S&P Global Commodity Insights for the period October 2024–January 2025; supplier financial reporting where publicly available; internal procurement performance records maintained by the Strategic Procurement team; and intelligence gathered during supplier engagement sessions conducted in January and February 2025.

1.3 Regulatory Obligations

The organization is subject to the monitoring, reporting, and strategic stock obligations applicable to downstream users under the CRMA. The relevant obligations for the current assessment period are as follows. Under Article 19, the organization is required to participate in the Commission's annual supply chain stress test for the strategic raw materials in its procurement portfolio. The Q4 2024 stress test was completed in January 2025, and the organization's submission is on file with the Legal Affairs Office under reference LAO-CRMA-2025-003. Under Article 20, the organization is required to submit an annual report on its strategic raw materials procurement, supply chain exposure, and diversification progress to the competent national authority by 30 April 2025.

Preparation of this submission is currently underway and will draw directly on the findings of this report. Under Article 8, the organization is required to maintain strategic stocks of critical raw materials at a level sufficient to cover 30 days of normal production. Compliance status for each material is addressed in Section 5 of this report.

2. Material Risk Profiles

2.1 Battery-Grade Lithium Carbonate

Risk classification: MEDIUM Composite risk score: 3.4 / 5.0

Supply concentration for battery-grade lithium carbonate has decreased compared to the previous assessment period, primarily due to capacity expansions in Australia and the ramp-up of new production in Argentina. The three-country HHI (Herfindahl-Hirschman Index) for refined lithium carbonate production currently stands at approximately 0.28, down from 0.34 in the Q2 2024 assessment. While this represents a meaningful improvement in the supply concentration dimension, the organization's own procurement exposure remains concentrated: approximately 79% of its lithium carbonate requirements are currently sourced from a single supplier under the framework agreement expiring on 30 June 2025.

Price volatility for lithium carbonate has been significant over the assessment period. Spot prices declined from approximately USD 13,800/t in October 2024 to a low of USD 11,200/t in January 2025, representing a 19% decline in a single quarter. The Directorate assesses that this price level is below the long-run marginal cost of production for a significant portion of global supply, and that a price correction is probable in H2 2025 as high-cost producers curtail output. This assessment underpins the recommendation to lock in volume under a long-term framework agreement during the Q2 negotiating round while market conditions remain favorable to buyers.

The regulatory and trade policy risk dimension is scored at 2.0 for lithium carbonate, reflecting the relatively stable policy environment in the principal producing jurisdictions. Chile and Argentina have both concluded Strategic Partnership Agreements with the European Union under Article 35 of the CRMA, which provides a degree of supply security assurance that is not available for other materials in the portfolio. The principal regulatory risk for this material is the potential tightening of Chilean lithium governance following the Chilean government's 2023 National Lithium Strategy, which introduced state participation requirements for new lithium production concessions. The organization's preferred supplier, Atacama Norte Mining S.A., operates under an existing concession and is not directly affected by the new requirements, but the medium-term supply landscape in Chile may be affected by reduced private investment in new capacity.

The assessment of the responsible contact for lithium carbonate procurement and associated supplier account management is:

[CONFIDENTIAL] Full name: Rodrigo Sepúlveda Mora | RUT: 13.876.542-7 | Title: Senior Supply Chain Analyst — Battery Materials | Email: r.sepulveda@strategic-supply.internal | Direct line: +56 2

2345 6801 | Assigned supplier accounts: Atacama Norte Mining S.A.; Pacific Lithium Holdings Ltd. | Annual procurement authority limit: USD 8,500,000 [/CONFIDENTIAL]

2.2 Neodymium

Risk classification: HIGH Composite risk score: 4.3 / 5.0

Neodymium presents the highest composite risk score of the four materials assessed in this report. The supply concentration dimension is scored at 5.0, reflecting the fact that approximately 85% of global refined neodymium production originates from a single jurisdiction, with no strategic partnership agreement in force between that jurisdiction and the European Union under the CRMA framework. The European Commission's Q4 2024 stress test identified neodymium as one of five materials with a critical short-term disruption probability, citing recent changes to export licensing procedures introduced by the principal producing jurisdiction in September 2024. These changes have not yet materially affected the organization's supply, but have introduced lead time uncertainty for Q2 2025 deliveries from the current spot supplier.

Price volatility for neodymium has been moderate during the assessment period. Prices have remained in the range of USD 82,000–94,000/t since October 2024, with no significant directional trend. The Directorate does not assess a near-term price spike as the primary risk scenario for this material; the primary risk is a volume availability disruption arising from export restriction escalation. In a severe disruption scenario modeled by the Directorate's risk team, a 60-day supply interruption would exhaust the organization's current strategic stock within 47 days and would require emergency procurement at estimated spot premiums of 25–40% over normal contract prices.

The EU-based preferred supplier, Rare Earth Supply GmbH, has a production facility in Lower Saxony that sources neodymium oxide from a diversified input supplier base and processes it to the specification required by the organization. Transitioning the majority of neodymium procurement to Rare Earth Supply GmbH would materially reduce the supply concentration risk for this material, at the cost of a moderate price premium estimated at approximately 6–8% over current spot market prices. This transition is the central strategic recommendation of this assessment for the neodymium category and underpins the negotiating position established for the Q2 round.

[CONFIDENTIAL] Internal risk scenario — RESTRICTED: Neodymium worst-case disruption model | Trigger event: export ban or quota restriction by primary producing jurisdiction | Estimated duration: 90 days | Strategic stock coverage at current consumption rate: 47 days | Estimated emergency procurement cost: USD 12,400,000 | Budget reserve activated: yes | Reserve account: CRM-CONTINGENCY-2025-ND | Authorized access: COO, CFO, Director of Strategic Procurement [/CONFIDENTIAL]

2.3 Cobalt Sulphate

Risk classification: MEDIUM-LOW Composite risk score: 2.7 / 5.0

The supply concentration risk for cobalt sulphate has decreased meaningfully since the Q2 2024 assessment, primarily as a result of the partial onshoring of cobalt refining capacity within the

European Union. Two EU-based refiners — including the organization's incumbent supplier, EuroCobalt Refining S.A. — now supply approximately 31% of the EU downstream market for battery-grade cobalt sulphate, up from 18% two years ago. This structural change reduces the overall supply concentration score for the material, though it does not fully eliminate upstream dependency on the Democratic Republic of Congo (DRC) as the dominant source of mined cobalt ore, which accounts for approximately 72% of global mine production.

The principal risk for cobalt sulphate in the current period is not supply concentration but supplier quality performance. EuroCobalt Refining S.A.'s incoming inspection rejection rate of 3.2% over the past two quarters, noted in the Executive Summary, represents a significant deterioration from the 0.8% rate recorded over the preceding four quarters. The Directorate's quality team has identified the root cause as an inconsistency in the sulphur content of the cobalt sulphate produced by EuroCobalt's secondary feedstock processing line, which was introduced to the supplier's production process in August 2024. EuroCobalt has acknowledged the issue and submitted a corrective action plan, but the plan has not yet been independently verified.

The supplier quality risk is addressed in the Q2 negotiating round through a targeted renegotiation of EuroCobalt's quality assurance clauses, as described in the memorandum SDG-CRM-2025-014. The Directorate's recommendation is to retain EuroCobalt as the primary cobalt sulphate supplier pending verification of the corrective action plan, while simultaneously accelerating the qualification of African Cobalt Resources Ltd. as a secondary supplier. Qualification activities for African Cobalt Resources Ltd. are currently in progress, with a target completion date of 31 July 2025.

2.4 Natural Graphite (Spherical, Anode Grade)

Risk classification: HIGH Composite risk score: 4.1 / 5.0

Natural graphite is assessed as HIGH risk for the current period, driven by a combination of near-critical supply concentration and an elevated regulatory and trade policy risk score. Virtually all spherical anode-grade natural graphite entering global supply chains undergoes processing in a single jurisdiction, which accounts for approximately 91% of global processing capacity for this material stage. Recent policy developments in that jurisdiction have introduced uncertainty regarding the continuity of graphite exports, including the introduction of export permit requirements for graphite products effective December 2023, which have increased lead times and reduced supply predictability for EU downstream users.

The organization's current natural graphite procurement is sourced on a spot basis from two EU-based distributors, Continental Graphite GmbH and Graphite Ventures S.A.S., both of which source processed graphite from the same upstream jurisdiction. The current arrangement provides no meaningful supply chain diversification at the processing stage. The Directorate's risk model identifies this as the organization's single highest-concentration upstream dependency across all materials in the Tier 1 portfolio.

The synthetic graphite qualification programme currently underway is the medium-term structural mitigation for this risk. However, as noted in the Executive Summary, the qualification process is not expected to be completed before Q4 2025. In the interim, the Directorate recommends securing a

12-month bridge supply agreement for natural graphite during the Q2 round, at current market rates, with an explicit contractual right to transition to synthetic graphite substitutes without volume penalty upon completion of qualification. This clause is a non-negotiable element of the Q2 negotiating position for this material.

The technical lead for the synthetic graphite qualification programme is:

[CONFIDENTIAL] Full name: Isabel Ferreira Andrade | RUT: 17.452.310-K | Title: Principal Materials Engineer — Anode Technologies | Email: i.ferreira@strategic-supply.internal | Lab access credentials — qualification testing system: user: iferreira_qual | Server: 10.0.5.14 | Port: 9443 | Project classification: CONFIDENTIAL — MATERIALS SUBSTITUTION [/CONFIDENTIAL]

3. Comparative Risk Summary

The following table presents the composite risk scores and dimension-level scores for the four materials assessed in this report. Scores reflect the current assessment period (Q4 2024–Q1 2025). For reference, the previous period scores are included in parentheses.

Material	SC	PV	RT	SF	SU	Composite	Rating	Previous
Lithium carbonate	3.5 (4.0)	3.0 (2.5)	2.0 (2.0)	2.5 (2.5)	4.5 (4.5)	3.4 (3.6)	MEDIUM	MEDIUM
Neodymium	5.0 (5.0)	2.5 (2.5)	5.0 (4.0)	3.0 (3.0)	4.5 (4.5)	4.3 (4.0)	HIGH	HIGH
Cobalt sulphate	2.5 (3.0)	2.0 (2.0)	2.0 (2.0)	3.5 (2.0)	3.0 (3.0)	2.7 (2.5)	MED-LOW	MED-LOW
Natural graphite	5.0 (5.0)	2.0 (2.0)	4.5 (3.5)	2.5 (2.5)	4.0 (4.0)	4.1 (3.7)	HIGH	MEDIUM

SC = Supply concentration; PV = Price volatility; RT = Regulatory/trade policy risk; SF = Supplier financial/operational stability; SU = Substitutability. Weights applied per RAP v4.1. Previous period scores in parentheses.

4. Strategic Stock Compliance Status

Article 8 of the CRMA requires the organization to maintain strategic stocks of critical raw materials at a level sufficient to cover 30 days of normal production. The following table presents the current strategic stock status as of the date of this report.

Material	30-day requirement	Current stock	Coverage (days)	Compliance status

Lithium carbonate	41.7 t	53.2 t	38.3	COMPLIANT
Neodymium	9.9 t	7.8 t	23.6	NON-COMPLIANT
Cobalt sulphate	18.3 t	22.1 t	36.2	COMPLIANT
Natural graphite	28.4 t	19.6 t	20.7	NON-COMPLIANT

The non-compliance status for neodymium and natural graphite requires immediate remediation action. The Directorate recommends emergency stock replenishment purchases for both materials within the next 30 days, funded from the CRM Strategic Stock Reserve budget line. Authorization for emergency stock purchases above EUR 500,000 requires written approval from the Chief Financial Officer. The estimated cost of bringing both materials to full compliance is EUR 1,840,000 for neodymium and EUR 620,000 for natural graphite.

[CONFIDENTIAL] Emergency stock replenishment authorization — RESTRICTED | Neodymium: approved amount EUR 1,840,000 | Budget line: CRM-STOCK-RESERVE-2025-ND | Authorized by: CFO — approval ref. CFO-2025-0089 | Natural graphite: approved amount EUR 620,000 | Budget line: CRM-STOCK-RESERVE-2025-GR | Authorized by: CFO — approval ref. CFO-2025-0091 | Transfer account: IBAN ES76 0049 1234 5678 9012 3456 | Counterpart bank: Banco Santander Corporate — Madrid [/CONFIDENTIAL]

5. Recommendations

The Directorate submits the following twelve recommendations to the Critical Resources Management Committee for consideration and decision at the March 2025 ordinary session.

Procurement strategy: R1. Authorize the Q2 negotiating round as described in SDG-CRM-2025-014, with the reserve positions and authorization levels specified therein. R2. Approve the transition of neodymium procurement to a long-term framework agreement with Rare Earth Supply GmbH as primary supplier, targeting a contract signature by 30 June 2025. R3. Approve emergency stock replenishment purchases for neodymium (EUR 1,840,000) and natural graphite (EUR 620,000) as detailed in Section 4, subject to CFO countersignature. R4. Authorize the continuation of the synthetic graphite qualification programme with a target completion date of 31 October 2025 and a budget allocation of EUR 380,000 for the remainder of the qualification process.

Supplier management: R5. Initiate formal quality dispute proceedings with EuroCobalt Refining S.A. in accordance with the dispute resolution clause of the current supply agreement, and establish a 90-day corrective action verification period. R6. Accelerate the qualification of African Cobalt Resources Ltd. as secondary cobalt sulphate supplier, with a target qualification completion date of 31 July 2025. R7. Initiate a Tier 2 supplier qualification process for a second neodymium supplier

from a jurisdiction covered by a CRMA Strategic Partnership Agreement, targeting qualification by Q4 2025.

Regulatory compliance: R8. Assign responsibility for preparation of the CRMA Article 20 annual report to the Legal Affairs Office, with input from the Strategic Supply Directorate, targeting submission by 15 April 2025. R9. Review and update the organization's Strategic Raw Materials Plan to reflect the findings of this assessment, with particular attention to the elevated risk rating for natural graphite. R10. Commission an independent verification of EuroCobalt Refining S.A.'s corrective action plan by a qualified third-party auditor, with results to be reported to the Committee by 30 June 2025.

Risk monitoring: R11. Establish a monthly supply disruption monitoring dashboard for neodymium and natural graphite, reporting directly to the Chief Operating Officer, effective from 1 April 2025. R12. Conduct an unscheduled interim risk assessment for neodymium and natural graphite in August 2025, in advance of the regularly scheduled biannual assessment, to assess whether the HIGH risk classification has been maintained or reduced following implementation of the recommendations above.

6. Document Control

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This report was prepared by the Risk Analysis Unit of the Strategic Supply Directorate and reviewed by the Chief Operating Officer and the Legal Affairs Office prior to distribution. Queries regarding its content should be directed to the Director of Strategic Procurement.